

Disproof of Conjecture 00000003497

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Statement and verdict

The conjecture asserts a one-to-one correspondence between Hermitian metrics and holomorphic structures through their Chern connections. As stated, it is false: distinct metrics can induce the same Chern connection.

Counterexample

Let $E = \mathbb{C}$ be the trivial holomorphic line bundle over a point. Define two Hermitian metrics by

$$h_1(z, w) = z\bar{w}, \quad h_2(z, w) = 2z\bar{w}.$$

They are different because $h_1(1, 1) = 1$ and $h_2(1, 1) = 2$. A connection has no directional data on a point, so there is exactly one connection and both metrics induce it. Consequently the metric-to-Chern-connection map is not injective and cannot be bijective.

The witness also works on a trivial holomorphic line bundle over any connected complex manifold. In the standard holomorphic frame the Chern connection form is $A = h^{-1}\partial h$. Both $h = 1$ and $h = 2$ are constant, hence $\partial h = 0$ and $A = 0$ for both.

Proposition 1. *The claimed one-to-one correspondence is false.*

Proof. The two metrics above are unequal and have equal Chern connection. Therefore the correspondence is not injective. Every bijection is injective, so it is not a bijection. $\square$

Scope

The familiar uniqueness of the Chern connection says that, for a *fixed* holomorphic structure and a *fixed* Hermitian metric, there is a unique compatible connection. The counterexample does not dispute that theorem; it refutes the stronger injectivity assertion actually filed.

The accompanying Lean project formalizes the finite core of this witness: positive scales 1 and 2 are distinct, the connection type on a point is a singleton, and the induced map is not injective.